

SMALL OSCILLATIONS (Chapter 6 Goldstein)

ID CASE: $\ddot{q} = f(q)$ EQUILIBRIUM $\Leftrightarrow 0 = f(q_0)$
at $q = q_0$

Now we want to know what happens if we move a little bit away from q_0

$q = q_0 + \eta$ η "SMALL"

$(q_0 + \eta)'' = \ddot{\eta}$ $f(q_0 + \eta) = \cancel{f(q_0)} + \eta f'(q_0) + \dots$

$\ddot{\eta} \approx f'(q_0) \eta$ SMALL DEVIATION

(EQUATION OF MOTION IS **LINEAR** IN SMALL DEVIATIONS)

OSCILLATIONS $\Leftrightarrow f'(q_0) < 0$

$\omega = \sqrt{-f'(q_0)}$

$\eta = \eta_0 \cos(\omega t + \varphi_0)$

INSTABILITY $\Leftrightarrow f'(q_0) > 0$

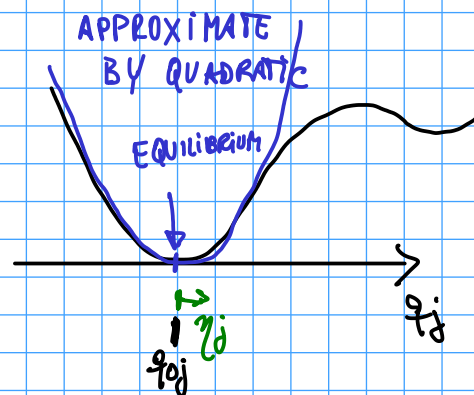
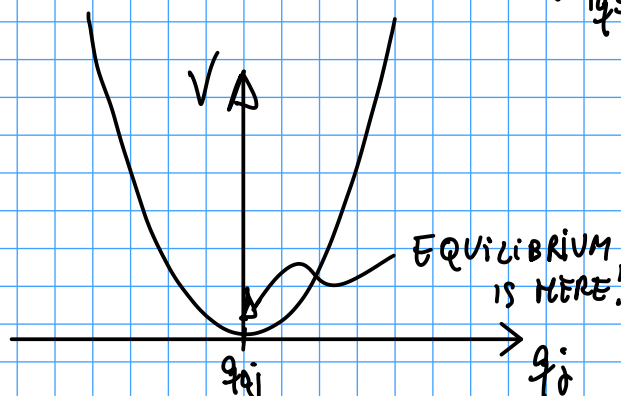
$\lambda = \sqrt{f'(q_0)}$

$\eta = A e^{\lambda t} + B e^{-\lambda t}$

More than one degree of freedom (Simplify by assuming $V = V(q_1, \dots, q_n, t)$)

$L = T - V$

Equilibrium at $(q_1, \dots, q_n) = (q_{01}, \dots, q_{0n}) \Leftrightarrow$ GENERALIZED FORCES $Q_j = - \frac{\partial V}{\partial q_j} \Big|_{\vec{q} = \vec{q}_0} = 0$



TAYLOR EXPAND V TO 2nd ORDER IN η

$V(q_1, \dots, q_n) = V(q_{01}, \dots, q_{0n}) + \underbrace{\sum_{j=1}^n \frac{\partial V}{\partial q_j} \Big|_{\vec{q}_0}}_{\substack{\text{CONSTANT} \\ \Rightarrow \text{NO WORRIES} \\ 0}} \eta_j + \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2 V}{\partial q_i \partial q_j} \Big|_{\vec{q}_0} \eta_i \eta_j + \dots$

$q_j = q_{0j} + \eta_j$

$\equiv V_{ij}$ $n \times n$ MATRIX $V_{ij} = V_{ji}$

$T = \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2 T}{\partial \dot{q}_i \partial \dot{q}_j} \Big|_{\substack{\vec{q} = \vec{q}_0 \\ \dot{\vec{q}} = \vec{0}}} \dot{q}_i \dot{q}_j$

$T_{ij} = T_{ji}$

$\vec{q} = \vec{q}_0$
 $\dot{\vec{q}} = \vec{0}$

$T = \sum_{p=1}^N \frac{m_p}{2} \left(\dot{\vec{r}}_p \right)^2$ ASSUME

$\dot{\vec{r}}_p = \sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t}$

$$L = \frac{1}{2} \sum_{i,j=1}^n (T_{ij} \dot{q}_i \dot{q}_j - V_{ij} q_i q_j)$$

$$\frac{\partial}{\partial q_2} \left(\frac{1}{2} [V_{11} q_1 q_1 + V_{12} q_1 q_2 + V_{21} q_2 q_1 + V_{22} q_2 q_2] \right) = \frac{1}{2} [(V_{12} + V_{21}) q_1 + 2 V_{22} q_2] \stackrel{V_{12}=V_{21}}{=} V_{21} q_1 + V_{22} q_2$$

($\frac{1}{2}$ PREFACTOR CANCELS WITH FACTOR OF 2 IN THE DERIVATIVE)

SIMPLE CASE $T = \frac{m}{2} \dot{q}^2$ $V = \frac{k}{2} q^2$
 $L = \frac{m}{2} \dot{q}^2 - \frac{k}{2} q^2$ $\frac{\partial L}{\partial q} = -kq$ $\frac{\partial L}{\partial \dot{q}} = m\dot{q}$
 $0 = \frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = -kq - m\ddot{q} \Rightarrow \boxed{m\ddot{q} = -kq}$

In general:

$$\frac{\partial L}{\partial q_i} = - \sum_{j=1}^n V_{ij} q_j \quad p_i = \frac{\partial L}{\partial \dot{q}_i} = \sum_{j=1}^n T_{ij} \dot{q}_j$$

Lagrange
Eqs:

$$0 = \frac{\partial L}{\partial q_i} - \dot{p}_i \Rightarrow \boxed{0 = \sum_{j=1}^n -V_{ij} q_j - \sum_{j=1}^n T_{ij} \ddot{q}_j} \quad i=1, \dots, n$$

Propose: $q_j = q_{j0} \cos(\omega t + \delta_j) \Rightarrow \ddot{q}_j = -\omega^2 q_j \quad i=1, \dots, n$

$$0 = \sum_{j=1}^n (V_{ij} q_j - \omega^2 T_{ij} q_j) = \sum_{j=1}^n (V_{ij} - \omega^2 T_{ij}) q_{j0} \cos(\omega t + \delta_j) \quad i=1, \dots, n$$

BUT ALSO, since $\cos(\omega t + \delta_j) \neq 0$ for most times:

$$0 = \sum_{j=1}^n (V_{ij} - \omega^2 T_{ij}) q_{j0} \quad i=1, \dots, n$$

EIGENVALUE

EIGENVECTOR

ω : NORMAL
MODE FREQUENCY

q_{j0} : NORMAL
MODE

For the q_{j0} to be nontrivial, we need

$$\boxed{\text{Det} [V_{ij} - \omega^2 T_{ij}] = 0}$$

ASIDE:

More details on the Kinetic Energy part:

$$T = \sum_{p=1}^N \frac{m_p}{2} (\dot{\vec{r}}_p)^2 = \sum_{p=1}^N \frac{m_p}{2} \left(\sum_{i=1}^n \frac{\partial \vec{r}_p}{\partial q_i} \dot{q}_i + \frac{\partial \vec{r}_p}{\partial t} \right) \cdot \left(\sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t} \right)$$

$$\vec{r}_p = \vec{r}_p(q_1, \dots, q_n, t) \Rightarrow \dot{\vec{r}}_p = \sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t}$$

$$T = T_2 + T_1 + T_0$$

$$T_2 = \frac{1}{2} \sum_{i,j=1}^n m_{ij} \dot{q}_i \dot{q}_j$$

$$m_{ij} = \sum_{p=1}^N m_p \frac{\partial \vec{r}_p}{\partial q_i} \cdot \frac{\partial \vec{r}_p}{\partial q_j}$$

(m_{ij} is a function of $(q_1, \dots, q_n)$
because the derivatives $\frac{\partial \vec{r}_p}{\partial q_k}$
depend on $(q_1, \dots, q_n)$)

$$T_1 = \sum_{j=1}^n \pi_j \dot{q}_j$$

$$\pi_j = \sum_{p=1}^N m_p \frac{\partial \vec{r}_p}{\partial q_j} \cdot \frac{\partial \vec{r}_p}{\partial t}$$

$$T_0 = \sum_{p=1}^N \frac{m_p}{2} \left(\frac{\partial \vec{r}_p}{\partial t} \right)^2$$

Assuming $\frac{\partial \vec{p}}{\partial t} = \vec{0}$, and expanding for $q_j = q_{0j} + \eta_j$ with η_j small, gives

$$T = T_2 = \frac{1}{2} \sum_{i,j=1}^n \left(m_{ij} \Big|_{\vec{q}=\vec{q}_0} + \sum_k \frac{\partial m_{ij}}{\partial q_k} \Big|_{\vec{q}=\vec{q}_0} \eta_k + \dots \right) (q_{0i} + \eta_i) (q_{0j} + \eta_j)$$

But $(q_{0i} + \eta_i)' = \dot{\eta}_i$ AND $\eta_k \dot{\eta}_i \dot{\eta}_j \propto \eta^3$ IS OF HIGHER ORDER IN SMALL DEVIATIONS

THEREFORE,
UP TO QUADRATIC
ORDER IN $\eta, \dot{\eta}$:

$$T = \frac{1}{2} \sum_{i,j=1}^n T_{ij} \dot{\eta}_i \dot{\eta}_j \quad \text{WITH} \quad T_{ij} \equiv m_{ij} \Big|_{\vec{q}=\vec{q}_0}$$

$$\text{Det} [V_{ij} - \omega_k^2 T_{ij}] = 0 \quad k=1, \dots, n \quad n \text{ \# of degrees of freedom}$$

$$[V - \omega_k^2 T] \vec{a}_k = \vec{0} \quad k=1, \dots, n$$

General solution is linear combination:

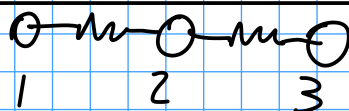
$$\eta_i = \sum_k C_k a_{ik} \cos(\omega_k t + \delta_k) \quad \vec{\eta} = \sum_k C_k \vec{a}_k \cos(\omega_k t + \delta_k)$$

Define $(A)_{ij} \equiv a_{ij} \quad A \equiv (\vec{a}_1 \vec{a}_2 \dots \vec{a}_n) \quad \underline{\text{Property:}} \quad 1 = A^T T A \Rightarrow A^{-1} = A^T T$

$$A^T V A = \begin{pmatrix} \omega_1^2 & 0 & \dots & 0 \\ 0 & \ddots & & \\ 0 & & \omega_n^2 & 0 \end{pmatrix}$$

Matrix of eigenvector / normal modes

Normal modes example:



$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$\circ \rightarrow \circ \leftarrow \circ$$

ω_1

$$\vec{a}_2 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

$$\leftarrow \circ \rightarrow \leftarrow \circ$$

ω_2

$$\vec{a}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\circ \rightarrow \circ \rightarrow \circ$$

$\omega_3 = 0$

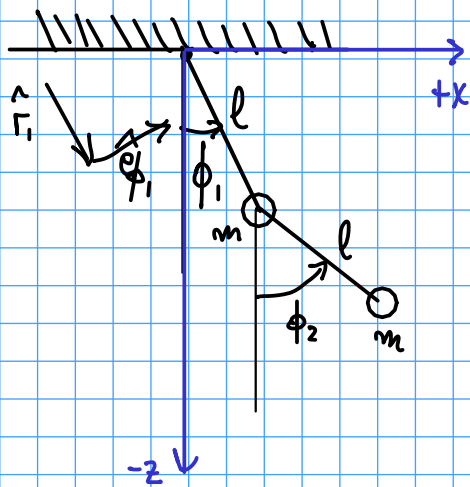
$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = C_1 \vec{a}_1 \cos(\omega_1 t + \delta_1) + C_2 \vec{a}_2 \cos(\omega_2 t + \delta_2) + C_3 \vec{a}_3 \cos(\omega_3 t + \delta_3)$$

Normal coordinates $\vec{\xi}$: $\vec{\eta} = A \vec{\xi} \quad \eta_i = \sum_k \xi_k a_{ik} \quad \xi_k = C_k \cos(\omega_k t + \delta_k) \quad \vec{\xi} = A^{-1} \vec{\eta} = A^T T \vec{\eta}$

$$L = \frac{1}{2} \sum_{i,j} (\dot{\eta}_i T_{ij} \dot{\eta}_j - \eta_i V_{ij} \eta_j) = \frac{1}{2} \left[\left(\frac{d\vec{\eta}}{dt} \right)^T T \frac{d\vec{\eta}}{dt} - \vec{\eta}^T V \vec{\eta} \right] = \frac{1}{2} \left[\vec{\xi}^T A^T T A \vec{\xi} - \vec{\xi}^T A^T V A \vec{\xi} \right]$$

$$L = \frac{1}{2} \left[\left(\frac{d\vec{\xi}}{dt} \right)^T \vec{\xi} - \vec{\xi}^T \begin{pmatrix} \omega_1^2 & 0 \\ 0 & \ddots & \omega_n^2 \end{pmatrix} \vec{\xi} \right] \Rightarrow \boxed{L = \frac{1}{2} \sum_{k=1}^n (\dot{\xi}_k^2 - \omega_k^2 \xi_k^2)}$$

Example: Oscillations of a double pendulum



Transformation equations

$$\vec{r}_1 = l (\sin \phi_1 \hat{x} - \cos \phi_1 \hat{z}) = l \hat{r}_1$$

$$\vec{r}_2 = l (\sin \phi_1 \hat{x} - \cos \phi_1 \hat{z}) + l (\sin \phi_2 \hat{x} - \cos \phi_2 \hat{z}) = l(\hat{r}_1 + \hat{r}_2)$$

$$\hat{r}_1 = \sin \phi_1 \hat{x} - \cos \phi_1 \hat{z} \quad \hat{r}_2 = \sin \phi_2 \hat{x} - \cos \phi_2 \hat{z}$$

Velocities

$$\dot{\vec{r}}_1 = \dot{\phi}_1 \frac{\partial \hat{r}_1}{\partial \phi_1}$$

$$\frac{\partial \hat{r}_1}{\partial \phi_1} = \hat{e}_{\phi_1} = \cos \phi_1 \hat{x} + \sin \phi_1 \hat{z}$$

$$\dot{\vec{r}}_2 = \dot{\phi}_2 \frac{\partial \hat{r}_2}{\partial \phi_2}$$

$$\frac{\partial \hat{r}_2}{\partial \phi_2} = \hat{e}_{\phi_2} = \cos \phi_2 \hat{x} + \sin \phi_2 \hat{z}$$

$$\dot{\vec{r}}_1 = l \dot{\phi}_1 (\cos \phi_1 \hat{x} + \sin \phi_1 \hat{z})$$

$$\dot{\vec{r}}_2 = l(\dot{\phi}_1 (\cos \phi_1 \hat{x} + \sin \phi_1 \hat{z}) + \dot{\phi}_2 (\cos \phi_2 \hat{x} + \sin \phi_2 \hat{z}))$$

$$T = \frac{m}{2} \left[(\dot{\vec{r}}_1)^2 + (\dot{\vec{r}}_2)^2 \right] = \frac{m}{2} \left[l^2 \dot{\phi}_1^2 + l^2 (\dot{\phi}_1^2 + \dot{\phi}_2^2 + 2 \dot{\phi}_1 \dot{\phi}_2 \hat{e}_{\phi_1} \cdot \hat{e}_{\phi_2}) \right] \quad \hat{e}_{\phi_1} \cdot \hat{e}_{\phi_2} = \cos \phi_1 \cos \phi_2 + \sin \phi_1 \sin \phi_2 = \cos(\phi_2 - \phi_1)$$

$$T = \frac{m l^2}{2} [2 \dot{\phi}_1^2 + \dot{\phi}_2^2 + 2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_2 - \phi_1)] \quad V = m g z_1 + m g z_2 = -m g l (2 \cos \phi_1 + \cos \phi_2)$$

$$L = T - V = \frac{m l^2}{2} [2 \dot{\phi}_1^2 + \dot{\phi}_2^2 + 2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_2 - \phi_1)] + m g l (2 \cos \phi_1 + \cos \phi_2)$$

We know $\phi_1 = \phi_2 = 0$ is stable equilibrium position, but let's check

$$\frac{\partial V}{\partial \phi_1} = m g l \sin \phi_1 = 0 \text{ at } \phi_1 = 0 \quad \frac{\partial V}{\partial \phi_2} = m g l \sin \phi_2 = 0 \text{ at } \phi_2 = 0$$

And V is absolute minimum at $\phi_1 = \phi_2 = 0 \Rightarrow$ Equil is STABLE

We thus expand T and V to quadratic order in $\eta_i = \phi_i - \phi_i^0 = \phi_i$

$$\phi_1 \phi_2 \cos(\phi_2 - \phi_1) = \eta_1 \eta_2 \cos(\eta_2 - \eta_1) \approx \eta_1 \eta_2 \left[1 - \frac{(\eta_2 - \eta_1)^2}{2} + \dots \right] \approx \eta_1 \eta_2 + O(\eta^4)$$

$$\cos \phi_i = \cos \eta_i = 1 - \frac{\eta_i^2}{2} + \frac{\eta_i^4}{24} - \dots \approx 1 - \frac{\eta_i^2}{2}$$

$$T = \frac{1}{2} (\dot{\eta}_1 \dots \dot{\eta}_n) \Pi \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_n \end{pmatrix}$$

Thus, to quadratic order in η and $\dot{\eta}$ we get

$$V = -m g l \left[3 - \frac{\eta_1^2}{2} - \frac{\eta_2^2}{2} \right] \Rightarrow V = m g l \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{CHECK}$$

$$(\eta_1 \ \eta_2) \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = (\eta_1 \ \eta_2) \begin{pmatrix} 2\eta_1 \\ \eta_2 \end{pmatrix} = 2\eta_1^2 + \eta_2^2 \quad \checkmark$$

$$T = \frac{m l^2}{2} [2 \dot{\eta}_1^2 + \dot{\eta}_2^2 + 2 \dot{\eta}_1 \dot{\eta}_2] \Rightarrow T = m l^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

$$(\dot{\eta}_1 \ \dot{\eta}_2) \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \dot{\eta}_1 \\ \dot{\eta}_2 \end{pmatrix} = (\dot{\eta}_1 \ \dot{\eta}_2) \begin{pmatrix} 2\dot{\eta}_1 + \dot{\eta}_2 \\ \dot{\eta}_1 + \dot{\eta}_2 \end{pmatrix} = 2\dot{\eta}_1^2 + 2\dot{\eta}_1 \dot{\eta}_2 + \dot{\eta}_2^2 \quad \checkmark$$

To simplify the algebra, we write $mg\ell = m\ell^2\left(\frac{g}{\ell}\right)$ $\omega_0^2 = g/\ell$

Eigenvalue equation: $0 = \text{Det}[V - \omega^2 \Pi] = \text{Det}\left(m\ell^2 \begin{bmatrix} 2g/\ell - 2\omega^2 & -\omega^2 \\ -\omega^2 & g/\ell - \omega^2 \end{bmatrix}\right)$

$$\Leftrightarrow 0 = (2\omega_0^2 - 2\omega^2)(\omega_0^2 - \omega^2) - \omega^4 = 2\omega^4 - 4\omega^2\omega_0^2 + 2\omega_0^4 - \omega^4 = \omega^4 - 4\omega_0^2\omega^2 + 2\omega_0^4$$

$$\Leftrightarrow 0 = (\omega^2 - 2\omega_0^2)^2 - \underbrace{(4\omega_0^4 - 2\omega_0^4)}_{2\omega_0^4} \Leftrightarrow \omega^2 = 2\omega_0^2 \pm \sqrt{2}\omega_0^2 = (2 \pm \sqrt{2})\omega_0^2 \Leftrightarrow \omega = \begin{cases} \omega_+ = \sqrt{2+\sqrt{2}}\omega_0 \\ \omega_- = \sqrt{2-\sqrt{2}}\omega_0 \end{cases}$$

COMPLETE SQUARES $\nearrow$

Eigenfrequencies

Eigenvector:

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\omega_0^2 - 2\omega^2 & -\omega^2 \\ -\omega^2 & \omega_0^2 - \omega^2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \Leftrightarrow \begin{cases} 0 = 2(\omega_0^2 - \omega^2)a_1 - \omega^2 a_2 \Leftrightarrow a_2 = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right)a_1 \Leftrightarrow \frac{a_2}{a_1} = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) \\ 0 = -\omega^2 a_1 + (\omega_0^2 - \omega^2)a_2 \Leftrightarrow a_1 = \left(\frac{\omega_0^2}{\omega^2} - 1\right)a_2 \Leftrightarrow \frac{a_2}{a_1} = \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1\right)} \end{cases}$$

SIDE NOTE: The 2 equations are equivalent $\Leftrightarrow 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) = \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1\right)} \Leftrightarrow 1 = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right)\left(\frac{\omega_0^2}{\omega^2} - 1\right)$

$$\Leftrightarrow \omega^4 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) \Leftrightarrow 0 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) - \omega^4 \Leftrightarrow 0 = \text{Det}(V - \omega^2 \Pi) \quad \checkmark$$

So we can take either equation and use it, together with normalization, to get $\vec{a}$:

Propose $\vec{a} = N \begin{pmatrix} 1 \\ 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) \end{pmatrix}$ N : normalization constant $x = \frac{\omega^2}{\omega_0^2}$

$$1 = [\vec{a}]^T \Pi [\vec{a}] = N^2 \begin{pmatrix} 1 & 2(x-1) \end{pmatrix} m\ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2(x-1) \end{pmatrix} = N^2 m\ell^2 \begin{pmatrix} 1 & 2(x-1) \end{pmatrix} \begin{pmatrix} 2x \\ 2x-1 \end{pmatrix}$$

$$1 = N^2 m\ell^2 [2x + 2(x-1)(2x-1)] = 2N^2 m\ell^2 [2x^2 - 3x + x + 1] = 2N^2 m\ell^2 [2x^2 - 2x + 1]$$

$$1 = 2N^2 m\ell^2 [2(x-1)^2 + 4x - 2 + 1] = 2N^2 m\ell^2 \underbrace{[2(x-1)^2 - 1]}_0 + 2x = 4N^2 m\ell^2 \frac{\omega_0^2}{\omega^2} \Rightarrow \boxed{N_{\pm} = \frac{1}{\sqrt{4m}} \frac{\omega_{\pm}}{\ell\omega_0} = \left(\frac{2 \pm \sqrt{2}}{4m\ell^2}\right)^{1/2}}$$

$$(x-1)^2 = x^2 - 2x + 1 \quad x^2 = (x-1)^2 + 2x - 1$$

$$2x^2 - 2x + 1 = 2[(x-1)^2 + 2x - 1] - 2x + 1 = [2(x-1)^2 - 1] + 2x$$

$$\omega^4 [2(x-1)^2 - 1] = [2(x\omega^2 - \omega_0^2)^2 - \omega^4] = [2(\omega_0^2 - \omega^2)^2 - \omega^4] = 0$$

$$\frac{\omega_0^2}{\omega_{\pm}^2} = \frac{1}{2 \pm \sqrt{2}} = \frac{2 \mp \sqrt{2}}{4 - 2} = 1 \mp \frac{1}{\sqrt{2}} \quad 2\left[\frac{\omega_0^2}{\omega_{\pm}^2} - 1\right] = \mp \sqrt{2}$$

$$\boxed{\vec{a}_{\pm} = \left(\frac{2 \pm \sqrt{2}}{4m\ell^2}\right)^{1/2} \begin{pmatrix} 1 \\ \mp \sqrt{2} \end{pmatrix}} \Leftrightarrow \boxed{\omega_{\pm} = (2 \pm \sqrt{2})^{1/2} \frac{g}{\ell}}$$

Modal

matrix:

$$A = (\vec{a}_+ \vec{a}_-) = (4m\ell^2)^{1/2} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

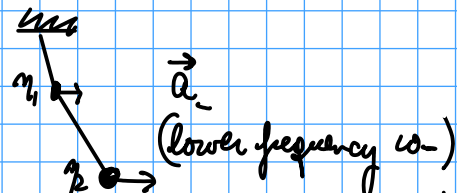
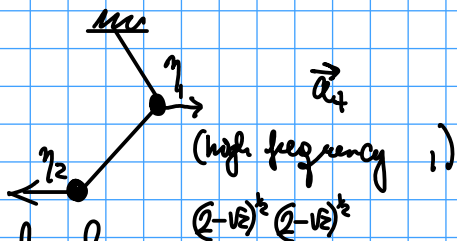
$$a = (2 + \sqrt{2})^{1/2} = -c/\sqrt{2}$$

$$c = -\sqrt{2}a = -\sqrt{2}(2 + \sqrt{2})^{1/2}$$

$$b = (2 - \sqrt{2})^{1/2} = d/\sqrt{2}$$

$$d = +\sqrt{2}b = +\sqrt{2}(2 - \sqrt{2})^{1/2}$$

Graphical representation:



$(\eta_2 = \sqrt{2}|\eta_1| > |\eta_1|)$
 IN BOTH MODES

Normal coordinates:

$$\vec{q} = A^T \Pi \vec{\eta} = A^T m l^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$

$$\vec{q} = (4 m l^2)^{1/2} \begin{pmatrix} a & c \\ b & d \end{pmatrix} m l^2 \begin{pmatrix} 2\eta_1 + \eta_2 \\ \eta_1 + \eta_2 \end{pmatrix} = \left(\frac{m l^2}{4}\right)^{1/2} \begin{pmatrix} (2a+c)\eta_1 + (a+c)\eta_2 \\ (2b+d)\eta_1 + (b+d)\eta_2 \end{pmatrix}$$

$$\begin{aligned} 2a+c &= (2-k)a = \sqrt{2}(\frac{k}{2(1-k)})^{1/2}(\frac{k}{2(1-k)})^{1/2}(2+k)^{1/2} = \sqrt{2} \frac{k}{2} (\frac{k}{2(1-k)})^{1/2} \\ a+c &= (1-k)a = -(\frac{k}{2(1-k)})^{1/2}(\frac{k}{2(1-k)})^{1/2}(2+k)^{1/2} = -\frac{k}{2} (\frac{k}{2(1-k)})^{1/2} \\ 2b+d &= (2+k)b = \sqrt{2}(\frac{k}{2(1+k)})^{1/2}(\frac{k}{2(1+k)})^{1/2}(2-k)^{1/2} = \sqrt{2} \frac{k}{2} (\frac{k}{2(1+k)})^{1/2} \\ b+d &= (1+k)b = (\frac{k}{2(1+k)})^{1/2}(\frac{k}{2(1+k)})^{1/2}(2-k)^{1/2} = \frac{k}{2} (\frac{k}{2(1+k)})^{1/2} \end{aligned}$$

$$\left(\frac{m l^2}{4}\right)^{1/2} = m^{1/2} \frac{l}{2}$$

$$\frac{1}{2} \frac{k}{2} (\frac{k}{2(1+k)})^{1/2} = \left(\frac{k l^2}{2(1+k)}\right)^{1/2}$$

$$q_+(t) = m^{1/2} l \left(\frac{k}{2(1-k)}\right)^{1/2} [\sqrt{2}\eta_1(t) - \eta_2(t)]$$

$$q_-(t) = m^{1/2} l \left(\frac{k}{2(1+k)}\right)^{1/2} [\sqrt{2}\eta_1(t) + \eta_2(t)]$$

But we know also that:

$$q_+(t) = F_+ \cos(\omega_+ t + \delta_+)$$

$$q_-(t) = F_- \cos(\omega_- t + \delta_-)$$

$$\dot{q}_+(t) = m^{1/2} l \left(\frac{k}{2(1-k)}\right)^{1/2} [\sqrt{2}\dot{\eta}_1(t) - \dot{\eta}_2(t)]$$

$$\dot{q}_-(t) = m^{1/2} l \left(\frac{k}{2(1+k)}\right)^{1/2} [\sqrt{2}\dot{\eta}_1(t) + \dot{\eta}_2(t)]$$

$$\dot{q}_+(t) = -F_+ \omega_+ \sin(\omega_+ t + \delta_+)$$

$$\dot{q}_-(t) = -F_- \omega_- \sin(\omega_- t + \delta_-)$$

Lagrangian in terms of the normal coordinates:

$$L = \frac{1}{2} \sum_{\alpha} (\dot{q}_{\alpha}^2 - \omega_{\alpha}^2 q_{\alpha}^2) = \frac{1}{2} [(\dot{q}_+^2 - \omega_+^2 q_+^2) + (\dot{q}_-^2 - \omega_-^2 q_-^2)]$$

$$L = \frac{1}{2} \left[\left(\dot{q}_+^2 - \left[(2+k) \frac{g}{l} \right] q_+^2 \right) + \left(\dot{q}_-^2 - \left[(2-k) \frac{g}{l} \right] q_-^2 \right) \right]$$

Assume initial condition is that at $t=0$: $\phi_1 = \phi_2 = 0$ $\dot{\phi}_1 = -\dot{\phi}_2 = v > 0$

Get $\phi_1(t)$, $\phi_2(t)$

Then

$$F_+ \cos(\delta_+) = q_+(0) = 0 \Rightarrow \delta_+ = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$$

$$F_- \cos(\delta_-) = q_-(0) = 0 \Rightarrow \delta_- = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$$

$$-F_+ \omega_+ \sin(\delta_+) = \dot{q}_+(0) = m^{1/2} l \left(\frac{k}{2(1-k)}\right)^{1/2} [\sqrt{2}v - (-v)] = \sqrt{2+1/k} \frac{m^{1/2} l v}{2}$$

$$-F_- \omega_- \sin(\delta_-) = \dot{q}_-(0) = m^{1/2} l \left(\frac{k}{2(1+k)}\right)^{1/2} [\sqrt{2}v + (-v)] = \sqrt{2-1/k} \frac{m^{1/2} l v}{2}$$

Choose $\delta_+ = \delta_- = -\frac{\pi}{2}$

$$F_{\pm} \sqrt{2 \pm 1/k} \sqrt{\frac{g}{l}} = \sqrt{2 \pm 1/k} \frac{m^{1/2} l v}{2} \Rightarrow F_{\pm} = \frac{v}{2} \left[\frac{m l^3}{g} \right]^{1/2}$$

$$\cos(\omega t - \frac{\pi}{2}) = \cos \omega t \cos \frac{\pi}{2} + \sin \omega t \sin \frac{\pi}{2} = \sin \omega t$$

$$\varphi_{\pm}(t) = f_{\pm} \cos(\omega_{\pm} t - \frac{\pi}{2}) = f_{\pm} \sin(\omega_{\pm} t)$$

$$\boxed{\varphi_{\pm}(t) = \frac{v}{2} \left[\frac{m l^3}{g} \right]^{\frac{1}{2}} \sin \left(\left[(2 \pm \sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right)}$$

In terms of the original coordinates

$$\vec{\eta}(t) = A \vec{\xi}(t) = (4 m l^2)^{\frac{1}{2}} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \varphi_+(t) \\ \varphi_-(t) \end{pmatrix} = (4 m l^2)^{\frac{1}{2}} \begin{pmatrix} a \varphi_+(t) + b \varphi_-(t) \\ c \varphi_+(t) + d \varphi_-(t) \end{pmatrix}$$

$$\begin{aligned} a &\equiv (2+\sqrt{2})^{\frac{1}{2}} = \frac{1}{\sqrt{2}}(\sqrt{2}+1)^{\frac{1}{2}} & c &= -\sqrt{2}a & f_+ a &= \frac{v}{2} \left[\frac{m l^3}{g} \right]^{\frac{1}{2}} (2+\sqrt{2})^{\frac{1}{2}} \\ b &\equiv (2-\sqrt{2})^{\frac{1}{2}} = \frac{1}{\sqrt{2}}(\sqrt{2}-1)^{\frac{1}{2}} & d &= +\sqrt{2}b & f_- b &= \frac{v}{2} \left[\frac{m l^3}{g} \right]^{\frac{1}{2}} (2-\sqrt{2})^{\frac{1}{2}} \\ (4 m l^2)^{\frac{1}{2}} \frac{v}{2} \left[\frac{m l^3}{g} \right]^{\frac{1}{2}} &= \frac{v}{4 \sqrt{2}} \end{aligned}$$

Check initial condition:

At $t=0$, all $\sin(\dots)$ terms are zero, so we have:

$$\phi_1 = \phi_2 = 0 \quad \checkmark$$

$$\begin{aligned} \phi_1(t) = \eta_1(t) &= \frac{v}{4 \sqrt{2}} \left\{ (2+\sqrt{2})^{\frac{1}{2}} \sin \left(\left[(2+\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) + (2-\sqrt{2})^{\frac{1}{2}} \sin \left(\left[(2-\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) \right\} \\ \phi_2(t) = \eta_2(t) &= \frac{\sqrt{2}v}{4 \sqrt{2}} \left\{ -(2+\sqrt{2})^{\frac{1}{2}} \sin \left(\left[(2+\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) + (2-\sqrt{2})^{\frac{1}{2}} \sin \left(\left[(2-\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) \right\} \end{aligned}$$

Check initial condition: At $t=0$, all $\cos(\dots)$ terms are 1, so we have:

$$\dot{\phi}_1 = \dot{\eta}_1(t) = \frac{v}{4} \left\{ (2+\sqrt{2}) \cos \left(\left[(2+\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) + (2-\sqrt{2}) \cos \left(\left[(2-\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) \right\}$$

$$\dot{\phi}_2 = \dot{\eta}_2(t) = \frac{\sqrt{2}v}{4} \left\{ -(2+\sqrt{2}) \cos \left(\left[(2+\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) + (2-\sqrt{2}) \cos \left(\left[(2-\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) \right\}$$

COMPARISON WITH QUANTUM MECHANICS:

Q M

$$|\psi(0)\rangle \rightarrow \{c_{\alpha}\}$$

$$|\psi(t)\rangle \leftarrow c_{\alpha} e^{-i E_{\alpha} t / \hbar}$$

$$|\psi(0)\rangle = \sum_{\alpha} c_{\alpha} |\varphi_{\alpha}\rangle \quad c_{\alpha} = \langle \varphi_{\alpha} | \psi(0) \rangle$$

$$|\psi(t)\rangle = \sum_{\alpha} c_{\alpha} e^{-i E_{\alpha} t / \hbar} |\varphi_{\alpha}\rangle$$

Small osc

$$\vec{\eta}(0), \dot{\vec{\eta}}(0) \longrightarrow \vec{\xi}(0), \dot{\vec{\xi}}(0)$$

$$\vec{\eta}(t), \dot{\vec{\eta}}(t) \longleftarrow \vec{\xi}(t), \dot{\vec{\xi}}(t)$$

$$\vec{\eta}(t) = A \vec{\xi}(t) \quad \vec{\xi}(t) = A^{-1} \vec{\eta}(t)$$